KINJAL GOSWAMI

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PROJECTS

SentinelAI — Production LLM Quality Monitoring System

June 2026

Live Demo · *Python · FastAPI · scikit-learn · Streamlit · Supabase (PostgreSQL)*

- Built a three-signal detection engine — **TF-IDF cosine similarity** for semantic drift, **Isolation Forest** for anomaly detection, and a **CUSUM statistical control algorithm** for gradual quality decay — to catch LLM failure modes a single metric would miss.
- Engineered a multi-tenant FastAPI backend with API-key authentication and automatic **per-tenant baseline generation** from live usage data, so each customer's "normal" is learned rather than hardcoded.
- Added a 50-sample warmup phase before detection activates, cutting false positives during cold starts, and shipped a Streamlit dashboard plus Slack alerting that flags degradation within 60 seconds of detection.
- Owned the full stack solo — architecture, **ML pipeline**, backend, deployment, and monitoring UI — with Supabase Postgres for persistence.

Contract Risk Analyzer — NLP Legal Contract Review Tool

August 2026

Live Demo · *Python · Hugging Face Transformers (DistilBERT) · Groq (Llama 3.3 70B) · Streamlit*

- Fine-tuned **DistilBERT** on the CUAD (Contract Understanding Atticus Dataset) to flag and classify clauses across 34 legal risk categories.
- Integrated **Groq's Llama 3.3 70B** to turn flagged clauses into plain-English risk explanations, so non-lawyers can understand what a contract actually commits them to.
- Handled the full pipeline solo — dataset preprocessing, **model fine-tuning and evaluation**, LLM prompt design, and frontend integration into a single upload-to-explanation tool.

Orbital Terrain Scanner — Satellite Land-Cover Classifier

September 2026

Live Demo · *Python · PyTorch (ResNet18) · Computer Vision · Rasterio · NumPy · Streamlit*

- Fine-tuned a pretrained **ResNet18** on the EuroSAT Sentinel-2 dataset, replacing the final classification layer to identify 10 land-cover classes (forest, water, industrial, residential, and others) from raw satellite imagery.
- Built a geospatial preprocessing pipeline with **rasterio and NumPy** to ingest .tif/.jpg satellite files and convert them into normalized tensors for model inference.
- Engineered an interactive anomaly filter — **NumPy matrix masking** over the model's classification output — that isolates exposed ground and infrastructure by excluding vegetation and water classes in real time.
- Shipped a Streamlit frontend with file upload, session-state handling, and custom CSS, and resolved Git buffer-size and timeout issues to deploy a 43MB model file to production.

OPEN SOURCE

HonmaruAI — Web Client Auth & Real-Time Sync

August 2026

Merged PR · *Cloudflare Workers*

- Turned the reference web client into a real signed-in client: session-token auth, a browser create/decide flow reaching the WebSocket relay, and a CORS fix that unblocked all browser API calls.
- Iterated through multiple rounds of maintainer review, closing authorization gaps the review surfaced (org-membership checks, invite-role validation, identity uniqueness) and adding a test for each fix before merge.

arkor — CLI Port-Fallback Fix

Merged PR

- Diagnosed and fixed a port-fallback bug in the `arkor dev` CLI command, taking the fix through multiple rounds of maintainer code review before it was merged into main.

TECHNICAL SKILLS

Languages: Python, C++, C, HTML/CSS

Frontend: Streamlit (interactive dashboards, session-state handling), custom CSS

Backend & APIs: FastAPI, REST APIs, PostgreSQL (Supabase), API-Key Auth, database design

AI / LLM Integration: Groq (Llama 3.3 70B) LLM integration, prompt design, Hugging Face Transformers fine-tuning, PyTorch, scikit-learn

Automation & Workflows: Slack Webhooks (automated alerting), automated per-tenant baseline generation, Isolation Forest & CUSUM anomaly workflows

Git, Testing & Deployment: Git, GitHub, unit testing (tests added per PR fix), Render deployment, NumPy, Pandas, Rasterio, Matplotlib/Seaborn, Jupyter, VS Code, DSA

ACHIEVEMENTS

- 100+ day coding streak on LeetCode; 150+ Data Structures & Algorithms problems solved.
- Awarded the Infosys STEM Stars Scholarship (2025–2026).
- Google Cloud Certificate: Introduction to Generative AI.
- Active on LinkedIn and X; writes technical build logs and lessons learned from ML/AI projects on X and Medium.